# 3CWZ for $CH_4$ and $NH_3$: progress against the protocol

Synthesis of the research protocol (O. Zaidi, CRNA), the 2nd- and 3rd-quarter 2026 scientific reports, and the code and calculations of this work (repository dramoros95/dd, folder nh3/).

## 1. Sources

| Source | Content | Role in this synthesis |
|---|---|---|
| Protocol (Protocole_ZAIDI_VF) | Objectives, methodology, 12-month plan, expected results | Reference checklist (sections 2–3) |
| Q2 report (2nd quarter 2026) | Molecular 3CWZ formalism: kinematics, TDCS, 3CWZ final state, Moccia OCE, Euler average (Eq. 19), $T_{dir} = T^1 - T^0$, Fourier transform of $C_{ab}$, split at r = d (Eqs. 23–36) | Theory of the amplitude |
| Q3 report (3rd quarter 2026) | $Z_{eff}(r)$ from the monopole Hartree potential of the Moccia density; self-interaction correction for the ejected electron; Z(r) curves for $CH_4$ and $NH_3$ | Theory of the variable charge |
| This work | $NH_3$ code (nh3_opt_claude.for), runs vs El Mir et al. (2020), short-range and orthogonalization tests, ejected-wave diagnosis, tests under Rezkallah et al. (2011) conditions | Implementation and results |

## 2. Protocol objectives: status

| Protocol objective (§4) | Where it is covered | Status | What remains |
|---|---|---|---|
| O1. Adapt 3CWZ to molecules with Moccia OCE orbitals; Z(r) for each MO | Q2 (formalism); Q3 (Z(r) derivation); code: Moccia Table I for $NH_3$ ($3a_1$, 1e, $2a_1$), ZSCR = Q3 Eq. (16), analytic orientation average; $CH_4$ $1t_2$ regression identical to the original code | DONE | Per-orbital decomposition of Z(r) (Q3 §4.3), optional |
| O2. TDCS for the orbitals of $CH_4$ and $NH_3$, several kinematics | $NH_3$: 3 orbitals at $E_s$ = 500 eV, $\theta_s = -6°$, $E_e$ = 37 and 74 eV; EMS at 1200 eV; DDCS at 500 eV. $CH_4$: $1t_2$ with the original code | PARTIAL | $CH_4$ $2a_1$ and new $CH_4$ runs with the same code; other kinematics |
| O3. PCI: compare results with and without $C_{12}$ | — | TO DO | Add a switch ($C_{12}$ = 1, $N_{ab}$ = 1) and rerun; cheap |
| O4. Validate against experiment and theory: $CH_4$ [35–37], $NH_3$ [38–39] | $NH_3$: El Mir et al. 2020 (protocol ref. [9]) at 74 eV; also Bawagan EMS and Opal DDCS | PARTIAL | Nixon et al. 2013 [39] (low energy, needs exchange); Rejoub et al. 2001 [38] (total CS, needs converged integrals); $CH_4$ Lahmam-Bennani 2009 [35] |
| Expected result: contribution of the variable charge | Z(r) computed and cross-checked; its effect on the TDCS not isolated yet | PARTIAL | Same runs with constant Z = 1 (3C/BBK limit) for a direct comparison |
| Expected result: validity domain of 3CWZ for molecules | Binary peak reproduced; recoil peak 5–10 times too small ($NH_3$, 74 eV); cause identified (ejected wave); numerical domain mapped (slow electrons, K < 3 a.u.) | PARTIAL | Extend to other kinematics; test the distorted-wave fix |

### Methodology steps (protocol §5)

| Step (planned duration) | Status | Comment |
|---|---|---|
| Bibliography (2 months) | PARTIAL | El Mir et al. 2020, Rezkallah et al. 2011, Moccia 1964 used; Champion et al. 2006 (1DW for $H_2O$) and Nixon et al. 2013 still to read |
| OCE orbitals and Z(r) in Fortran (1 month) | DONE | All MOs of Moccia Table I; norms = 1; Z(∞) = 1 (ion), 0 (neutral) |
| Molecular 3CWZ code (1 month) | DONE | Plus options isr (short-range potential) and iorth (orthogonalization) |
| Calculations $CH_4$ with PCI study (3 months) | TO DO | Only the regression test was run here |
| Calculations $NH_3$ vs experiment (3 months) | PARTIAL | 74 eV complete; 37 eV computed, data not digitized |
| Analysis and discussion (2 months) | PARTIAL | Recoil diagnosis and numerical limits (sections 5–6) |

## 3. The variable charge: Q3 report vs code

The Q3 note derives $Z_{eff}(r)$ as the monopole part of the Hartree potential of the Moccia density (its Eq. 16), with the ejected electron screened by one electron less (self-interaction correction, Eqs. 18–19). The code computes exactly this expression (function ZSCR). Values from the code:

| r (a.u.) | 0 | 0.5 | 1.0 | 1.5 | 2.0 | 2.5 | 3.0 | 4.0 |
|---|---|---|---|---|---|---|---|---|
| $NH_3$ incident (neutral) | 7.000 | 2.690 | 1.113 | 0.549 | 0.541 | 0.227 | 0.091 | 0.013 |
| $NH_3$ ejected ($3a_1$ hole) | 7.000 | 3.094 | 1.806 | 1.402 | 1.474 | 1.198 | 1.079 | 1.011 |

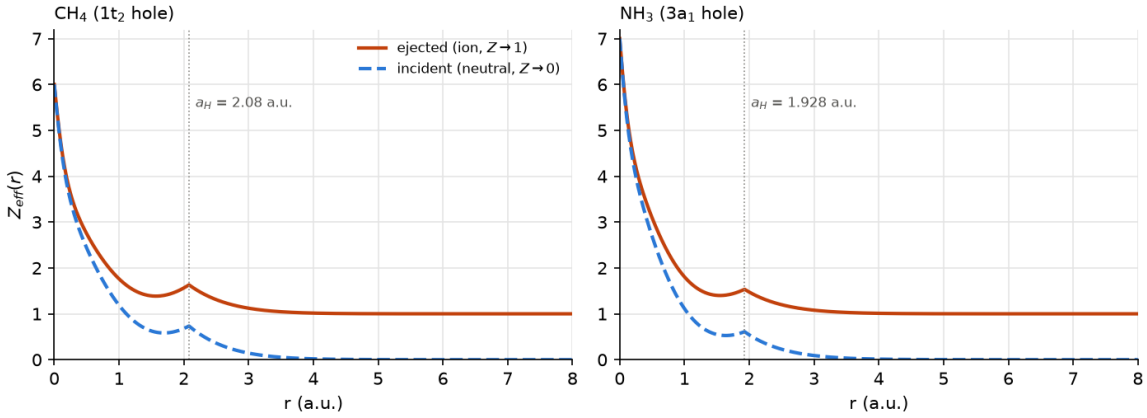



**Figure 1.** $Z_{eff}(r)$ from the code (files Zr_*_code.dat).

Every prediction of Q3 is confirmed by the code:

- $Z(0) = Z_N$ (6 for $CH_4$, 7 for $NH_3$).
- Asymptotes 0 (incident) and 1 (ejected) reached by r ≈ 4 a.u. ($Z_{ion}(4) = 1.011$). This also justifies the cut-off d = 4 a.u. of Q2 Eq. (34).
- $Z_{ej}(r) \geq Z_{inc}(r)$ at every r.
- Local maximum exactly at $r = a_H$ (1.928 a.u. for $NH_3$, 2.08 a.u. for $CH_4$): Z = 1.53 / 0.61 ($NH_3$ ejected / incident). In the shell model it is a kink, from $\min(r/a_H, 1)$.
- The three $NH_3$ orbitals give nearly the same $Z_{ion}(r)$ (within 0.05).

# 4. Results for $NH_3$ (this work)

## 4.1 TDCS vs El Mir et al. ($E_s$ = 500 eV, $\theta_s$ = −6°, $E_e$ = 74 eV)

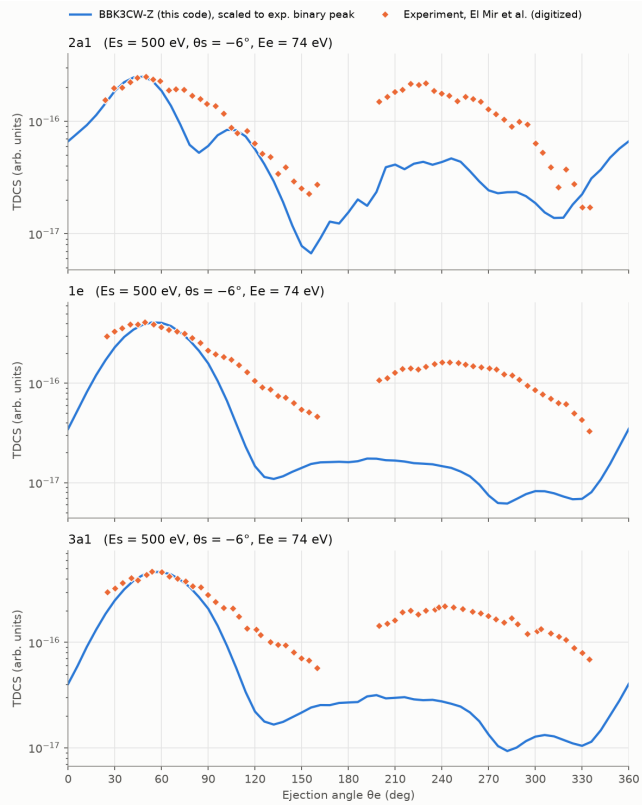



**Figure 2.** BBK3CW-Z (scaled to the experimental binary maximum) vs experiment. Detailed discussion: elmir_conditions/report/NH3_BBK3CWZ_vs_ElMir_Ee74eV_discussion.pdf.

| Orbital | Exp. binary / recoil | BBK3CW-Z | + short-range (isr = 1) | orthogonalized |
|---|---|---|---|---|
| $3a_1$ | 2.1 | 14.8 | 0.98 | 30 |
| 1e | 2.5 | 23.4 | 1.8 | 37 |
| $2a_1$ | 1.1 | 5.3 | 0.7 | 3.0 |

- **Binary peak**: position (48–60° vs 49–54°) and front side correct in every version.
- **Recoil peak and large-angle side**: 5–10 times too low for all three orbitals.
- **Short-range potential**: raises the recoil, but isotropically. Most of the rise comes from the non-orthogonality of the ejected wave; orthogonalization removes it and the recoil falls back.

## 4.2 Diagnosis: the ejected-electron wave

The ejected wave of Q2 Eq. (9) with the local α(r) = Z(r)/k does not solve Eq. (11) for the Q3 potential. The exact solution in the same $Z_{ion}(r)$ at 74 eV gives large short-range phase shifts, and the $NH_3^+$ core back-scatters far more than a Z = 1 field:

| l | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $\delta_l$ (rad) | 2.60 | 1.34 | 0.48 | 0.26 | 0.17 | 0.09 |

| elastic angle | 30° | 60° | 90° | 120° | 150° | 180° |
|---|---|---|---|---|---|---|
| exact / Rutherford (Z = 1) | 2.5 | 4.8 | 4.9 | 29 | 91 | 137 |

A plane-wave estimate confirms that the recoil cannot come from the orbitals: $\rho(q_{binary})/\rho(q_{recoil})$ = 38 ($3a_1$), 59 (1e), 155 ($2a_1$) against 1–2.5 measured. The recoil must come from the scattering of the ejected electron by the ion. Proposed fix: an exact distorted wave for the ejected electron in the Q3 potential (theory_distorted_wave.md).

### 4.3 Tests under the conditions of Rezkallah et al. (2011)

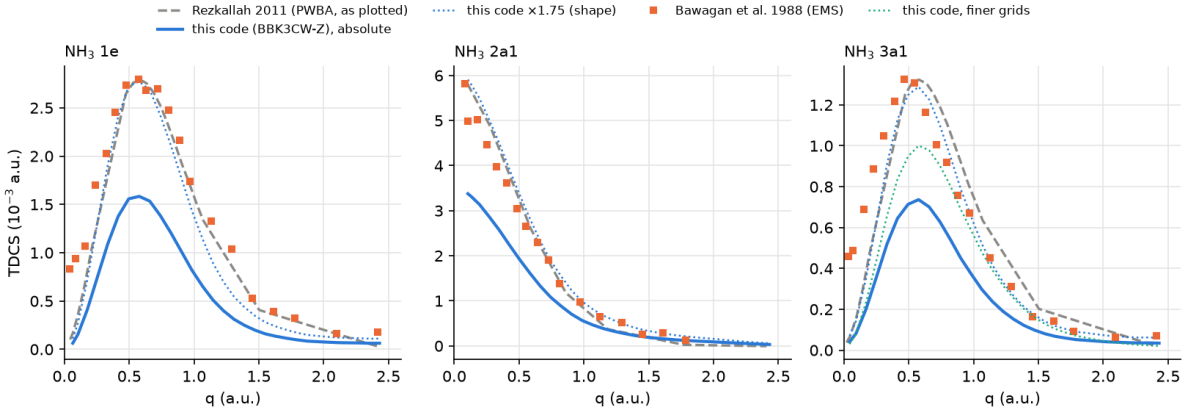



**Figure 3.** EMS momentum profiles (1200 eV, 45°, 600/600 eV) vs Bawagan et al. and the paper's PWBA.

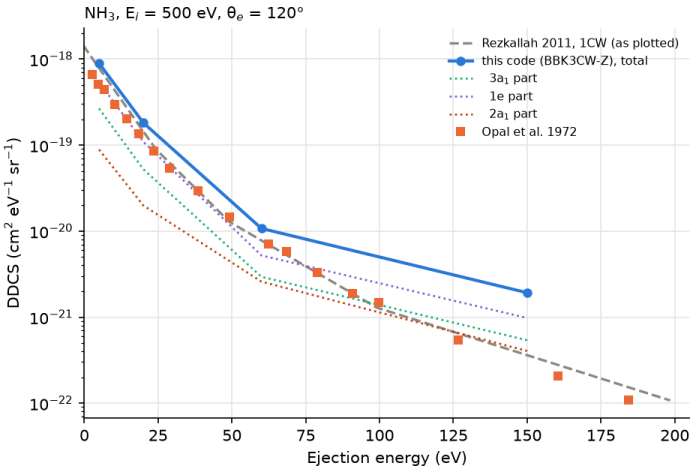



**Figure 4.** DDCS at $E_i$ = 500 eV, $\theta_e$ = 120° vs Opal et al.

- **EMS**: peak positions correct; shapes of $2a_1$ and $3a_1$ correct; the 1e high-momentum tail is ~1.5 times low (orbital). Absolute value 0.57 of the PWBA with default grids (0.76 with finer grids for $3a_1$).
- **DDCS**: model/experiment = 1.79, 1.58, 1.35 at $E_e$ = 5, 20, 60 eV; 7.1 at 150 eV, where the excess is numerical (large momentum transfer; finer grids lower it by 12–30 times).
- **Numerical domain**: the default grids (nleg = 10, nd = 4, md = 1) are reliable to ~20% only for slow outgoing electrons and K < 3 a.u. The El Mir conditions (K ≈ 0.85 a.u., 74 eV) are inside it.

## 5. Points to correct or clarify in the documents

| Document | Point | Suggested change |
|---|---|---|
| Q3, table §3 | $NH_3$ occupied MOs listed as "1A1, 2A1, 1A2, 1E" | Moccia Table I: $1a_1$, $2a_1$, $3a_1$, 1e (no $a_2$ orbital); $3a_1$ is the HOMO (lone pair) |
| Q3, table §3 and §4.2 | $a_H$(N–H) = 1.91 a.u. | Moccia Table I uses R = 1.928 a.u. (also in the code); use the value of the orbitals |
| Q3, §3.1 | "this is how PCI enters" | The +1 tail is the electron–ion final-state interaction; PCI is the electron–electron factor $C_{ab}$ (Q2 Eq. 13) |
| Q3, Figs. 1–2 | Colours of incident/ejected swapped between the two captions | Use the same convention |
| Q2, Eq. (2.25)/(22) | $\sigma \propto \lvert T_{dir}\rvert^2 + \lvert T_{exc}\rvert^2 + \lvert T_{dir} - T_{exc}\rvert^2$ | Check the factor ½ of the spin average: ½(...) = $\lvert T_d\rvert^2 + \lvert T_e\rvert^2 - \mathrm{Re}(T_d T_e^*)$ |
| Q2, Eq. (30) | Prefactor written $1/2\pi^3$ | Should read $1/(2\pi)^3$ as in Eq. (29) |
| Q2, Eq. (7) vs Q3 | Projectile–core potential $-1/r_0$ (frozen core), while the waves use Z(r) | State the choice; the full $-Z_{ion}(r)/r_0$ is option isr = 1 (tested, section 4.1) |

| Document | Point | Suggested change |
|---|---|---|
| Q2, §3.4 | Exchange amplitude dropped | Fine for $E_e$ = 74, $E_s$ = 500 eV; needed for Nixon et al. (equal energies) |
| Protocol, §2.3 | Claims better recoil peak with 3CWZ | Not confirmed for $NH_3$ at 74 eV; present it as a question the protocol tests |
| Protocol, ref. [26] and §5 | Moccia J. Chem. Phys. 40, 2164 (paper I) for both molecules; "Tables IV and IX" | Cite 2164 for $CH_4$ and 2176 (paper II, Table I) for $NH_3$, as Q3 does |
| Protocol, after Eq. (3) | $r_j$ called the "ejected electron" | $r_j$ are the target electrons |
| Protocol, objective 2 | "l'orbitale de $NH_3$" (singular) | The three valence orbitals are treated |
| Protocol, §5 | No numerical validation step | Add a grid-convergence step (section 4.3) |

## 6. Remaining work, ordered by cost and mapped to the protocol

| # | Task | Protocol item | Cost |
|---|---|---|---|
| 1 | PCI switch ($C_{12}$ = 1) and $NH_3$ runs at 74 eV | O3 | small |
| 2 | Same runs with constant Z = 1 (3C/BBK limit) to isolate the effect of Z(r) | Expected result 1 | small |
| 3 | Digitize the 37 eV data of El Mir et al. and compare (runs already done) | O4 | small |
| 4 | $CH_4$ runs with the same code ($1t_2$, $2a_1$) vs Lahmam-Bennani et al. 2009 [35] | O2, O4 | medium |
| 5 | Distorted wave for the ejected electron in the Q3 potential | Expected result 2, perspectives | large |
| 6 | Grid convergence / K-adapted grids, then integrated cross sections vs [38] | O4, method | medium |
| 7 | Exchange amplitude, then Nixon et al. 2013 [39] (low energy) | O4 | large |

## 7. Files (repository dramoros95/dd, branch claude/read-and-explain-k311k1)

| Path | Content |
|---|---|
| nh3/nh3_opt_claude.for | $NH_3$ 3CWZ code (options isr, iorth) |
| nh3/elmir_conditions/ | Runs, experimental data, figures and PDF discussion for El Mir et al. |
| nh3/theory_orthogonalization.md | Theory of the orthogonalization test |
| nh3/theory_distorted_wave.md, nh3/diagnostics/ | Ejected-wave diagnosis and proposed fix |
| nh3/rezkallah_conditions/ | EMS and DDCS tests, data, scripts, README |
| nh3/progress/ | This synthesis, Z(r) data and figure |